CHAPTER 6

Logarithm function

S44, S47, S49, S50

Before you start · exponential and inequities

Without a calculator, in six minutes. Respond briefly. Mobilize areas and diversion Composite. Then choose the targeted review indicated under the questions.

1. Resolve $e^x = 1$.

2. Give the sign of e^x.

3. Determine the area of $\frac{1}{x-3}$.

4. Derivating e^{2x}.

5. Resolve 2 $x - 1 > 0$.

6. Solve $e^x < e^2$.

Responses

$x = 0$; strictly positive ; $\mathbb{R} \setminus \{3\}$; $2e^{2x}$; $x > \frac{1}{2}$; $x < 2$.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Back to strictness exp growth and exp(0). **2.** Trace the exponential over the horizontal axis. **3.** Solve x-3=0 before taking the opposite. **4.** Name the inner function and its derivative; R36. **5.** Isolate x by two operations which keep order. **6.** Use strict exp growth to compare Exhibitors. Questions R serve as support for the teacher, even if their collective ritual is planned later.

Objectives

Use ln properties; solve equations and thresholds; drift and study functions Logarithmics.

Course notes

Definition and properties

The exponential is continuous, strictly increasing on $\mathbb{R}$, and his image is $]0, +\infty[$. For everything $x > 0$, there is a unique reality y such that $e^y = x$: we note it x.

$$\ln(e^y) = y \quad (y \in \mathbb{R}), \qquad e^{\ln x} = x \quad (x > 0), \qquad \ln 1 = 0, \quad \ln e = 1.$$

For $a, b > 0$ and $n \in \mathbb{Z}$,

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a, \quad \ln\sqrt{a} = \frac{1}{2}\ln a.$$

Product proof. $e^{\ln a+\ln b} = e^{\ln a}e^{\ln b} = ab$; the uniqueness of the antecedent gives the formula. no similar formula for ln $(a+b)$.

Derivative, Variations and Limits

We admit it's driftable on $]0,+\infty[$. Derivating $e^{\ln x} = x$ gives $e^{\ln x}(\ln x)' = 1$; therefore $(\ln x)' = \frac{1}{x} > 0$. Ln is strictly increasing and concave, because its second derivative is $-\frac{1}{x^2}$.

$$\lim_{x\to 0^+} \ln x = -\infty, \qquad \lim_{x\to+\infty} \ln x = +\infty.$$

Its curve is symmetrical with that of the exponential in relation to $y = x$. The right $x = 0$ is a vertical asymptote. Its tangent in 1 gives ln $x \leq x-1$ for $x > 0$.

Composition and Comparative Growth

If $u > 0$ is derivable, $(\ln u)' = \frac{u'}{u}$. For the integer $m \geq 1$,

$$\lim_{x\to+\infty} \frac{\ln x}{x^m} = 0, \qquad \lim_{x\to 0^+} x^m \ln x = 0.$$

Demonstration in 0. Lay down $t = -\ln x$; when $x \to 0^+$, $t \to +\infty$ and $x\ln x = -te^{-t} \to 0$ per comparison of growth rates. For $m \geq 1$, the same reasoning gives $-te^{-mt} \to 0$. In $+\infty$, pose $t = \ln x$ to write $\frac{\ln x}{x^m} = \frac{t}{e^{mt}} \to 0$.

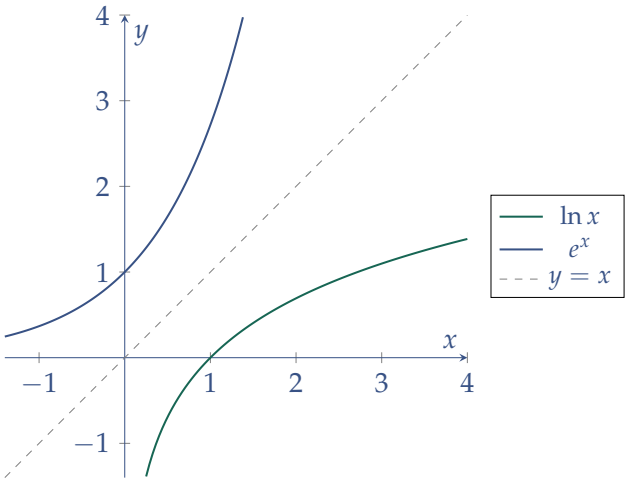
Equations and thresholds

For $A, B > 0$, $\ln A = \ln B \Leftrightarrow A = B$ and $\ln A \leq \ln B \Leftrightarrow A \leq B$. Always start with the domain.

Example. $\ln(x-1)+\ln(x+1) =$ No 8 requires $x > 1$. Equation becomes $x^2 - 1 = 8$, so $x =$ Three alone.

Threshold. $0{,}8^n \leq 0.1$ is equivalent to $n\ln 0{,}8 \leq 0{,}1$, then $n \geq \frac{\ln 0{,}1}{\ln 0{,}8} \simeq 10{,}319$. The first integer is 11.

Example of study. For $f(x) = \frac{\ln x}{x}$, $f' = \frac{1-\ln x}{x^2}$; f growth on $]0,e]$, decreases on $[e,+\infty[$, and its maximum value $\frac{1}{e}$. Its limit in 0 $^+$ is $-\infty$, its limit in $+\infty$ is 0.



Use logarithm with its conditions

6.1 Inverse of exponential

First use: S44.

Definition · Neperian logarithm

For $x > 0$, $\ln x$ is the unique real number y such that $e^y = x$. Thus $e^{\ln x} = x$ for $x > 0$ and $\ln(e^y) = y$ for all real y. The logarithm transforms a positive value into a real exponent.

Example Solved · Domain before Transformation

To solve $\ln(x-1) = \ln(3-x)$, the two arguments must be positive, therefore $1 < x < 3$. The strict growth of ln then gives $x - 1 = 3 - x$, therefore $x = 2$, which belongs to the domain.

A1 · Immediate application Identify areas of $\ln(2-x)$ and $\ln(x^2-1)$. Solve $\ln(2x+1) = 0$ by specifying the field.

6.2 Products, quotients and powers

First use: S46.

Property · Algebraic Rules

For $a, b > 0$, $\ln(ab) = \ln a + \ln b$ and $\ln(a/b) = \ln a - \ln b$. For a relative integer m, $\ln(a^m) = m \ln a$. The conditions on the arguments remain to be checked after a transformation.

Demonstration. Lay down $u = \ln a$, $v = \ln b$. Then $ab = e^u e^v = e^{u+v}$, therefore $(ab) = u + v$. The identity of the quotient follows from $\ln(b \times b^{-1}) = \ln 1 = 0$, and that of the powers by repetition of the product then passage On the other hand. □

Example solved · An exact expression

$\ln 12 - \ln 3 = \ln 4 = 2 \ln 2$. By contrast, $\ln(1+1) = \ln 2 \neq \ln 1 + \ln 1 = 0$. Logarithm not distributive on the addition.

A2 · Two errors to explain Refuse " $\ln(a+b) = \ln a + \ln b$ Then explain why $\ln(x^2) = 2 \ln x$ is not an identity on $\mathbb{R} \setminus \{0\}$ and give the correct identity on this set.

6.3 Studying a logarithmic function

First use: S47.

Example Solved · Variation and Maximum

On $]0; +\infty[$, $f(x) = \frac{\ln x}{x}$ a for derivative $f'(x) = \frac{1 - \ln x}{x^2}$. It grows up to $x = e$, then decreases, and its maximum is $1/e$. Its sign depends on $\ln x$, so it is negative before 1 ; this does not prevent its growth on the $]0; 1[$.

A3 · Partial guidance Study the domain, derivative and sign of $g(x) = \ln(1+x^2)$. Explain why the function is even and what it announces for variations.

6.4 The sign of $\ln q$ controls inequality

First use: S50.

Example Solved · A Lower Threshold

For $n \in \mathbb{N}$, $0{,}8^n < 0{,}1$ is equivalent to $n \ln 0{,}8 < \ln 0{,}1$. Like $\ln 0{,}8 < 0$, the inverse division Order: $n > \ln 0{,}1 / \ln 0{,}8 \simeq 10{,}32$. The first row is 11; $0{,}8^{10} \simeq 0{,}1074$ and $0{,}8^{11} \simeq 0{,}0859$ control the border. $q > 1$, $\ln q > 0$ and order is kept; for $q = 1$, we don't divides Not by ln $q = 0$.

A4 · Choosing common sense Determine the smallest integer $n \geq 0$ as 1 $1{,}05^n \geq 2$, then as $0{,}9^n \leq 0.2$. Check the previous rank in each case.

6.5 Compare and change variable

First use: S49.

Ownership · Common limitations

$\ln x \to +\infty$ in $+\infty$ and $\ln x \to -\infty$ in 0^+. For the integer $m \geq 1$, $(\ln x)/x^m \to 0$ in $+\infty$; of plus $x \ln x \to 0$ in 0^+. The first comparative growth is derived from $t/e^{mt} \to 0$ with $t = \ln x$.

Solved Example · Adapting Comparative Growth

In $+\infty$, $\frac{2+\ln x}{x} = 2/x + \ln x / x \to 0$. In 0^+, $x^2 \ln x$ becomes $-te^{-2t}$ with t=-ln x; this expression tend towards 0.

A5 · Deferred application Calculate the limits of $\ln(2x)/x$ in $+\infty$ and $x \ln(3x)$ in 0^+. To detail a transformation using algebraic rules.

A6 · Synthesis problem Depending on the real a, determine the number of positive $\ln x = ax$. Use ln study x/xits limits and the position of the level a.

A first step: Distinguish $a < 0$, $a = 0$, $0 < a < 1/e$, $a = 1/e$ and $a > 1/e$. An unmet limit does not provide not an additional solution.

Wait a minute at the courthouse. Explain why "taking logarithm" is not allowed on a expression whose positivity has not been verified.

Review postponed · S58 and then S66 Identify a cooling constant, then find a antiderivative $1/x$ keeping the definition interval.

Research activity

Control a calculation error · 30 min.

A calculation method produces an initial error $e_0 = 5$. At each step, the error is multiplied by 0.7. Note e_n error after n steps, for $n \in \mathbb{N}$.

1. Express e_n according to the n.

2. Using logarithm, determine the smallest integer n such that $e_n \leq 10^{-4}$.

3. Calculate, at the calculator, the error to the rank found and to the previous rank to check the minimality of the rank.

4. We now impose an error not more than 10^{-6}. Determine the new minimum number Does the division of tolerance by 100 multiply the number of steps per 100?

Progressive exercises

E1 · Direct application. After determining the set on which it is defined, resolve Equation $\ln(2x-1) = \ln(x+3)$.

E2 · Application and justification. Determining the real x for which $\ln x + \ln(x-2)$ is defined, then resolve $\ln x + \ln(x-2) = \ln 3$.

E3 · Method to be organised. It is considered $f(x) = \ln(x^2+1)$ and $g(x) = x\ln x$. Clarify their set and to calculate their derivative.

E4 · Choice of strategy. Determine $\lim\limits_{x\to+\infty} \dfrac{\ln x}{\sqrt{x}}$ and $\lim\limits_{x\to 0^+} x^2 \ln x$. Justifying growth results comparative employees.

E5 · Synthesis or demonstration. Using the study of a function, demonstrate that $\ln x \leq x-1$ for all real $x > 0$. The concavity of the logarithm shall not be used directly.

Work in the manual or annal

Barbazo, p. 196, Exercise 36. At home after S44: treating the three functions. before solving the inequities; justify the open bounds of the intervals obtained.

Duty at home

DM6 · Study of a logarithmic function · 55 min.

The function is considered f defined on $]0;+\infty[$ by $f(x) = \frac{1+\ln x}{x}$. Its representative curve is noted $\mathcal{C}_f$.

1. Determining the limits of f at the boundaries of the domain, study his sign, then draw his picture changes.

2. Study the convexity of f and determine the coordinates of the deflection point of $\mathcal{C}_f$.

3. Demonstrate that the equation $f(x) = \frac{1}{2}$ admits exactly two solutions. Give each one one justified framework using calculated values.

4. Using the maximum of f, demonstrate that $1 + \ln x \leq x$ for every $x > 0$. Specify the case equality.

QCM A · domain and properties

After S47; QCM B after S50.

For each question, only one proposal is accurate. consult the correction.

1. The domain of $\ln(2x-1)$ is:

A. $\mathbb{R}$ B. $[\frac{1}{2};+\infty[$

C. $]0;+\infty[$ D. $]\frac{1}{2};+\infty[$

2. For $a,b>0$, $\ln(ab)$ is valid for:

A. $\ln a+\ln b$ B. $\ln a \ln b$

C. $\ln(a+b)$ D. $a\ln b$

3. $\ln(e^{-3})$ is valid for:

A. 3 B. e^3

C. -3 D. $\ln 3$

4. The derivative of $\ln(1+x^2)$ is:

A. $\frac{1}{1+x^2}$ B. $\frac{2x}{1+x^2}$

C. $2\ln x$ D. $\frac{2}{x}$

5. The equation $\ln x = 2$ a for solution:

A. $2e$ B. $\ln 2$

C. 2 D. e^2

QCM B · Threshold, Growth and Parameter

For each question, only one proposal is accurate.
consult the correction.

6. The smallest integer $n \geq 0$ as $(\frac{1}{2})^n < 0.1$ is:

A. 3 B. 10

C. 4 D. 2

7. For $0 < q < 1$ and $0 < a < 1$, $q^n < a$ equals:

A. $n > \frac{\ln a}{\ln q}$ B. $n < \frac{\ln a}{\ln q}$

C. $n > \frac{\ln q}{\ln a}$ D. $n > a/q$

8. $\lim_{x\to+\infty} \frac{\ln x}{x}$ is valid for:

A. 1 B. 0

C. $+\infty$ D. $-\infty$

9. $\ln(x^2) = 2\ln x$ is valid:

A. for all real x B. for everything $x \neq 0$

C. only for $x > 1$ D. for everything $x > 0$

10. The maximum of $g(x) = \frac{\ln x}{x}$, $x > 0$, is:

A. 1 B. e

C. $1/e$ D. 0

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. $\ln(1+1) = \ln 1 + \ln 1$.

2. Divide inequality by ln 2 reverses its meaning.

3. $\ln x / x = 1$ has a positive solution.

Corrected MCQs

1. **D.** The argument must be strictly positive: 2 $x - 1 >$ Zero is excluded.

2. **A.** The logarithm transforms a product into a sum; no analogous rule transforms a product sum of arguments.

3. **C.** The exponential and logarithm functions are converse on their domains.

4. **B.** Apply u'/u with $u = 1 + x^2 > 0$; factor 2 x It's from the compound.

5. **D.** Apply the exponential to both members; $e^2 > 0$ checks the domain.

6. **C.** $2^{-3} = 0{,}125 \geq 0{,}1$ and $2^{-4} = 0{,}0625 < 0{,}1$. The continuation decreases, so no previous rank It's okay.

7. **A.** Take the logarithm then divide by ln $q < 0$ reverses the sense of inequality.

8. **B.** Comparative logarithm growth and positive logarithm power x.

9. **D.** The right member shall impose $x > 0$. For $x \neq 0$, the general identity is $(x^2) = 2\ln|x|$.

10. **C.** $g' = (1 - \ln x)/x^2$, positive before e then negative. The maximum reached in e is equal to $1/e$.

True/false. False: ln 2 $\neq 0$. False: ln 2 > 0. False: maximum 1/ $e < 1$ (question 10) prohibited.

Corrected applications and support

Application 1. Fields: $]-\infty;2[\ ;\ ]-\infty;-1[\cup]1;+\infty[$. For the equation, $x>-1/2$ and $2x+1=1$, where $x=0$.

Application 2. Take $a=b=1$. For $x=-1$, the left limb $\ln 1$ exists but $\ln(-1)$ does not exist not in the real. The good identity is $(x^2)=2\ln|x|$ for $x\neq 0$.

Implementation 3. Area $\mathbb{R}$, derivative $2x/(1+x^2)$, negative before 0 and positive after ; minimum $g(0)=0$, therefore $g\geq 0$. $g(-x)=g(x)$; symmetry confirms the opposite variations on both sides of 0.

Implementation 4. First threshold: $n\geq \ln 2/\ln 1{,}05\simeq 14{,}21$, so 15; $1{,}05^{14}\simeq 1{,}980$ and $1{,}05^{15}\simeq 2{,}079$. Second: $n\geq \ln 0{,}2/\ln 0{,}9\simeq 15{,}28$, therefore 16, $0.9^{15}\simeq 0.2059$ and $0.9^{16}\simeq 0{,}1853$.

Application 5. $\ln(2x)/x=\ln 2/x+\ln x/x\to 0$. $x\ln(3x)=x\ln 3+x\ln x\to 0$.

Implementation 6. $f(x)=\ln x/x$ tend towards $-\infty$ in 0^{+}, grows strictly up to $1/e$, then decreases Strictly around 0 remaining positive after e. Thus: a solution for $a<0$; one for $a=0$ (1) ; two for $0<a<1/e$; one for $a=1/e$ (e) ; none for $a>1/e$. The conclusions make use of the continuity on each branch.

Return on error

For the threshold, ask to first predict if the sequence increases or decreases, then compare This sense with the inequality obtained. To resume false distributivity with a product of which one 1. The parameter problem allows to work existence, singleness per branch and limits not reached.

Conduct and correction

Benchmarks for the teacher

S44 is placed before the first white, but the logarithm does not belong to its announced perimeter. Insisting on conditions $a > 0, b > 0$ before grouping logarithms. Differentiability of The derivation formula is shown for a threshold of 0. $< q < 1$, the sign negative ln q Inequality reverses. Control the candidate rank and the precedent.

Corrected activity

1. Each step multiplies the error by 0.7: $e_n = 5(0{,}7)^n$ for $n \geq 0$.

2. $e_n \leq 10^{-4}$ is equivalent to $(0{,}7)^n \leq 2 \times 10^{-5}$. By growth of ln, $n \ln 0{,}7 \leq \ln(2 \times 10^{-5})$. Like $\ln 0{,}7 < 0$, we get

$$n \geq \frac{\ln(2 \times 10^{-5})}{\ln 0{,}7} \simeq 30{,}33512.$$

   The first suitable integer is 31.

3. $e_{30} \simeq 1{,}1270 \times 10^{-4} > 10^{-4}$ and $e_{31} \simeq 7{,}8888 \times 10^{-5} < 10^{-4}$. The continuation of errors decreases, so this control confirms minimality.

4. For target 10^{-6} the same calculation gives $n \geq \frac{\ln(2\times10^{-7})}{\ln 0{,}7} \simeq 43{,}2465$, so $n = 44$. The actual threshold increased by $\frac{\ln 100}{|\ln 0{,}7|} \simeq 12{,}91$, then round to the next integer. The number of steps is not multiplied by 100.

E1. We need two. $x - 1 > 0$ and $x + 3 > 0$, therefore $x > \frac{1}{2}$. In this field, the focus of ln gives 2 $x - 1 = x + 3$, let $x = 4$. This value belongs to the domain; all solutions are $\{4\}$.

E2. We have to $x > 2$. The formula of the product gives $\ln(x(x-2)) = \ln 3$, so $x^2 - 2x - 3 = 0$, either $(x-3)(x+1) = 0$. Only $x = 3$ satisfied $x > 2$: the other root -1 is excluded.

E3. Since $x^2 + 1 > 0$, the first function is detrievable on $\mathbb{R}$ and its derivative is $\frac{2x}{x^2+1}$. On $]0, +\infty[$, the rule of the product gives $(x \ln x)' = \ln x + x(\frac{1}{x}) = 1 + \ln x$.

E4. For the first limit, positons $t = \sqrt{x} \to +\infty : \frac{\ln x}{\sqrt{x}} = \frac{2 \ln t}{t} \to 0$. For the second, positons $t = -\ln x \to +\infty : x^2 \ln x = -te^{-2t} \to$ These two limits use comparative growth.

E5. For $x > 0$, posisons $g(x) = x - 1 - \ln x$. We have $g'(x) = \frac{x-1}{x}$, negative before 1, positive after. minimum $g(1) = 0$, therefore $g(x) \geq 0$, which is equivalent to $x \leq x - 1$.

Corrected from DM6

1. In 0^+, the numerator tends to $-\infty$ and denominator to $0^+ : f \to -\infty$. In $+\infty$, $f = \frac{1}{x} + \frac{\ln x}{x} \to 0$. The sign is that of $1 + \ln x$: negative on $]0, e^{-1}[$, none in e^{-1}By quotient,

$$f'(x) = \frac{1 - (1 + \ln x)}{x^2} = -\frac{\ln x}{x^2}.$$

   So f grows strictly up to 1, then decreases strictly; its maximum is $f(1) = 1$.

2. By drifting $-(\ln x)x^{-2}$, we get $f''(x) = \frac{2 \ln x - 1}{x^3}$. The numerator changes sign, negative to the positive, in $x = \sqrt{e}$. So f is concave on $]0, \sqrt{e}]$, convex $[\sqrt{e}, +\infty[$ and has inflection $(\sqrt{e}, \frac{3}{2\sqrt{e}})$.

3. On $]0,1]$, continuity, strict growth, limit $-\infty$ and final value 1 give exactly one preimage of $\frac{1}{2}$. On $[1,+\infty[$, strict decrease from 1 to 0 gives exactly a second The first one is in $]e^{-1},\frac{1}{2}[$, since $f(e^{-1})=0$ and $f(\frac{1}{2})=2(1-\ln 2)\simeq 0{,}6137>\frac{1}{2}$. The second is in $]5,6[$, since $f(5)\simeq 0{,}5219>\frac{1}{2}>f(6)\simeq 0{,}4653$.

4. The maximum gives $\frac{1+\ln x}{x}\leq 1$. Like $x>0$, you can multiply without changing the meaning: $1+\ln x\leq x$. Equality is the only maximum of f, therefore $x=1$.

Corrected work in the manual or an anal

Logarithm imposes a strictly positive argument.

1. $3x-7>0\iff x>\frac{7}{3}$, therefore $D_f=]\frac{7}{3};+\infty[$.

2. $-x^2+4x-3=-(x-1)(x-3)>0$ exactly when $1<x<3$. Thus $D_g=]1;3[$.

3. It must be simultaneously $x>0$ and $2-x>0$; their intersection gives $D_h=]0;2[$. The coefficient -3 placed in front of the second logarithm does not change its definition condition.